

HELIOPATH – Prototype 1

Full Solar Corridor Infrastructure Model

Introduction

HELIOPATH is a smart solar infrastructure project developed with the aim of creating a reliable, cost-efficient, and sustainable energy solution for campus transportation pathways. The project mainly focuses on installing modular solar structures along footpaths and road pathways to generate clean electricity while also providing functional public infrastructure.

The increasing demand for electricity and the growing dependence on fossil fuels and coal-based energy systems have created the need for alternative renewable energy solutions. HELIOPATH was designed as a concept to support clean energy adoption and contribute toward sustainable infrastructure development. The project also aligns with India's long-term vision of reducing dependence on conventional fuel sources and moving toward cleaner energy systems by 2047.

The first prototype of HELIOPATH was developed as a large-scale solar corridor structure intended for full-road deployment. The main idea behind this prototype was to combine renewable energy generation with public utility infrastructure. The structure provides shaded pathways for pedestrians during hot summer conditions while simultaneously generating solar power through mounted solar panels.

In addition to solar energy generation, the prototype also integrates battery storage terminals for storing generated electricity and EV charging support for future electric mobility systems. The overall design was developed as a modular infrastructure concept capable of supporting green transportation and sustainable campus development.

Objective

The primary objective of Prototype 1 was to design a large-scale solar-powered infrastructure system that could:

- Generate renewable electricity using solar energy
- Provide shaded pathways and shelter during extreme weather conditions
- Support EV charging infrastructure
- Store generated energy using battery terminals
- Promote sustainable transportation systems
- Encourage clean and green mobility solutions

Another major objective of the prototype was to explore how existing road infrastructure could be converted into multifunctional smart infrastructure capable of serving both transportation and energy requirements simultaneously.

Prototype Description

Prototype 1 was designed as a large solar corridor structure with approximate dimensions of 6000 mm × 7000 mm. The structure was developed as a modular installation system covering a large road section. The prototype mainly focused on integrating solar infrastructure directly above transportation pathways.

The model contains a curved solar panel structure mounted over the pathway. The curved design was selected to improve structural appearance while also creating a futuristic smart-city style infrastructure model. The solar modules are mounted above the support structure and are intended to generate electrical energy from sunlight.

The prototype also includes a battery terminal system where generated electricity can be stored for later use. This stored energy can then be utilized for EV charging stations, lighting systems, or other infrastructure support systems.

To improve practicality, a drainage system was also integrated into the prototype. The drainage section was designed to help manage rainwater flow and improve overall infrastructure planning.

The prototype was mainly focused on full-road deployment instead of footpath-only implementation. Because of this, the structure became very large and required extensive structural coverage.

Features of Prototype 1

1. Curved Solar Structure

The prototype contains a curved solar roof system designed to support mounted solar panels while also providing shaded coverage.

2. Battery Storage Terminal

Battery terminals were integrated into the structure to store generated electrical energy for future usage.

3. EV Charging Support

The model supports EV charging infrastructure, promoting electric mobility and sustainable transportation systems.

4. Drainage System

A drainage section was added to improve practicality and infrastructure realism during rainy conditions.

5. Modular Design

The structure was designed using repeated modular sections, allowing future scalability and extension of the system.

6. Renewable Energy Integration

The prototype supports clean energy generation and contributes toward reducing dependence on fossil-fuel-based electricity systems.

Design and Development Process

The development of Prototype 1 involved multiple CAD iterations and structural redesigns. During the design process, several mistakes related to scaling, alignment, and structure integration were encountered. Multiple small prototype models were initially created before developing the final large-scale structure.

The curved solar structure required repeated adjustments to achieve proper alignment and spacing between modules. Different support structures and panel placements were tested to improve the appearance and stability of the model.

The prototype was developed as part of the learning process in CAD-based infrastructure design. Through repeated experimentation and redesign, the structure gradually evolved into a more organized and functional model.

Problems and Limitations

Although Prototype 1 successfully demonstrated the main concept of HELIOPATH, several limitations and design problems were identified during development.

1. Very Large Structure Size

The prototype was designed for full-road deployment, resulting in a very large structure that became difficult to manage and optimize.

2. Absence of Cycle Mounting System

The first prototype did not include a dedicated cycle mounting or parking section, limiting its support for sustainable cycling infrastructure.

3. Complex Structural Scaling

Due to the large dimensions of the model, maintaining consistent proportions and structural balance became difficult.

4. Practical Deployment Challenges

Installing solar structures over an entire road may require significant construction, cost, and maintenance resources.

Improvements Planned for Future Versions

Based on the limitations identified in Prototype 1, several improvements were planned for future prototypes:

- Transition from full-road coverage to footpath-based implementation
- Addition of cycle mounting and parking systems
- Improved structural proportions and realistic scaling
- Better support structures and cleaner road integration
- Enhanced infrastructure detailing and campus integration
- Improved CAD realism and modular arrangement

These improvements later influenced the development of Prototype 2, which focused more on campus-specific deployment and practical infrastructure planning.

Conclusion

Prototype 1 of HELIOPATH successfully established the foundational concept of integrating solar-powered infrastructure with transportation pathways. The prototype demonstrated the possibility of combining renewable energy generation, shaded pedestrian pathways, battery storage systems, and EV charging support within a single modular infrastructure system.

Although several structural and scaling limitations were identified, the prototype served as an important learning and development stage in the overall HELIOPATH project. It also helped improve CAD modeling skills, infrastructure planning understanding, and modular system design concepts.

The experience gained during the development of Prototype 1 created the foundation for more realistic and refined future prototypes focused on practical campus deployment and sustainable smart infrastructure systems.

